Home Automation Sign Language Recognition with application of MediaPipe Using Random Forest Algorithm

In Partial Fulfillment of the Requirements for the Course

Software Engineering

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August 2026



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Chapter I

THE PROBLEM AND ITS BACKGROUND

This chapter presents the rationale of the study by discussing the reasons behind the study and the goals to be achieved. In addition, the study identifies the beneficiaries and their benefits. Finally, the paper outlines the solution that can be implemented to address the problem and provides details on the sections and organization of this research. The chapter consists of such sections as introduction, background of the study, objectives of the study, significance of the study, scope and delimitation, conceptual framework of the study, and operational definition of terms.

1.1 Introduction

Sign language is a complete visual language that uses hand gestures, spatial position, movement trajectories, and facial expressions to convey meaning for Deaf and Hard-of-Hearing (DHH) individuals, it's their primary means of communication (Banag, 2023). Since it does not rely on auditory channels, sign language provides a straightforward way to interact with technology through computer vision.

Smart home automation systems make controlling household items such as lights, fans, and appliances easier. Making these systems compatible with sign language can benefit Deaf and Hard-of-Hearing (DHH) individuals by allowing them to operate it with a familiar and accessible modality. Instead of adapting to a new control scheme such as touchscreens, a sign-language smart interface can allow users to issue commands using gestures that they already know.



Recent developments in sign-language interfaces have demonstrated the potential of using visual and gesture-based interactions. Tran et al. (2024) found that the Deaf prefer using ASL to control household items such as lights and devices than using conventional touchscreen applications. The researchers' results emphasized the viability of using gesture control as a ubiquitous and contactless means of operating common devices.

Although smart home automation can provide many benefits for DHH individuals, the technology can be inaccessible due to its high cost. Hutchinson et al. (2024) reviewed the costs associated with home automation for individuals with disabilities. The researchers found that systems that control home lighting, temperature, and other appliances cost up to $17,777.69 and upwards of $37,000 when accounting for components, installation, and setup. This steep price puts the system out of reach for most families.

With recent advances in computer vision, executing sign language recognition is possible on common consumer-grade devices. Google's MediaPipe Hands technology can extract 21 three-dimensional hand landmarks from RGB videos without requiring specialized or high-end equipment (Zhang et al., 2020). Similar results were achieved by Shin et al. (2021) in their experiments, which demonstrated that hand landmarks can be used to recognize American Sign Language (ASL) while being memory efficient. Furthermore, studies have shown that using structured landmarks in sign language recognition can be accurate and computationally more efficient than using deep learning approaches (Gupta et al., 2024).

Many existing works of sign-language-based smart home automation have demonstrated the efficacy of MediaPipe in real-time applications. Yang et al. (2023) used MediaPipe's keypoints along with geometric calculations to develop a method of turning off and on various household appliances using a microcontroller. Similarly, Lopes et al. (2025) used a MediaPipe gesture controller that had satisfactory results in terms of latency and user feedback. As a result, a new lightweight and affordable approach is necessary to develop a real-time application that can be recognized by a consumer-grade webcam. Several existing works have used the MediaPipe ASL recognition model to develop real-time applications but have not focused on the core practical gestures of the language and h ave used more computationally expensive models such as convolutional neural networks (CNN), long short-term memory (LSTM), and vision transformers (VT).

This study aims to address these gaps by developing a real-time home automation system that uses a smartphone camera to capture selected ASL gestures, MediaPipe to extract hand landmarks, a Random Forest algorithm to classify those landmarks, and an Arduino-based relay interface to control household appliances. By concentrating on practical static gestures and emphasizing affordability and computational efficiency, the proposed system seeks to provide an accessible and cost-effective solution for the Deaf and Hard-of-Hearing community.

1.2 Related Review Literature

This section is designed to identify and enumerate several researches related to the present study. The purpose of this review is to set the current research project within a conceptual and theoretical context. This includes a number of citations from journals, case studies, theses, research papers and other reliable literature sources.

Agile And Scrum SDLC Methodologies

An objective and systematic approach is required for the development of the proposed system; thus, the Agile software development life cycle is deemed appropriate for this research. Studies have shown the limitations of traditional sequential models in the development of systems that incorporate machine learning, computer vision, and hardware. Projects with technical requirements, including the use of models training, accuracy testing, and hardware installation, experience shifting demands as their development nears completion. Research by Subramanian et al. (2021) and Bosch et al. (2021) showed that Waterfall and sequential approaches fail in artificial intelligence and computer vision due to the unpredictability of model training and real-time requirements. These rigid approaches assume that requirements are well-defined and stable at the initiation phase, which is rarely the case in data-driven projects. Iterative approaches, on the other hand, allow processes such as algorithm improvement and system testing to occur in cycles while the developers continue to adjust the requirements. Schwaber and Sutherland (2020) developed Scrum – a lightweight framework that promotes the iterative development of products and services. Their research highlighted the importance of short development sprints and continuous feedback in software development, where the



requirements keep changing, instead of long stretches of development that last until product launch.

Research in the field of Internet of Things (IoT) and smart home automation demonstrate similar trends. Akbar et al. (2022) and Silva et al. (2023) showed that Scrum allowed development teams to break their work into processes related to communication protocol identification between the applications and microcontrollers and test them iteratively. This approach allowed them to troubleshoot communication issues and address other problems in individual components without disrupting the rest of the system. By dividing the development process into discrete sprints, development teams were able to meet project requirements, including the simultaneous use of microcontrollers and real-time software instruction.

Similar trends were highlighted in academic and technical projects. Klopp et al. (2020), Schneider et al. (2020) and Hron and Obwegeser (2022) showed that Scrum could be adopted in academic and technical developments to promote teamwork, iterative development, and continuous feedback. Their findings showed that the framework allowed researchers to manage technical requirements effectively while also promoting project-specific feedback loops throughout development, ensuring that the project met the technical requirements at every stage. All three studies demonstrated that Scrum could be adopted to manage the development process of student and researcher projects that require a range of interrelated technical processes. All the above research highlights the need for iterative development in studies that incorporate machine learning, computer vision, and traditional hardware. The current study aims to develop a machine learning system that uses mobile

camera input and MediaPipe keypoint detection, Random Forest classification, and Arduino Uno R3 relay system. Therefore, this study proposes to adopt the Scrum framework to continue improving the processes and unit testing every aspect of the system in successive development sprints that lead to the launch.

Computer Vision and Machine Learning Pipelines

Deaf and Hard-of-Hearing (DHH) individuals rely on visual languages for daily communication, yet they continue to face significant barriers due to the limited number of people who understand sign language and the scarcity of human interpreters. The notes of Martinez and Cabalfin (Banag, 2023) explain that sign languages are fully visual languages that utilize hand movements, spatial placements, and facial expressions. Given this visual nature, computer vision is a suitable technology for developing non-verbal interaction systems. The present study builds on this foundation by using hand gestures as commands for controlling household appliances instead of relying on voice-based interfaces.

Grande et al. (2023) demonstrated that computer vision techniques can successfully recognize American Sign Language (ASL) hand gestures from camera-based input. Their system utilized OpenCV to process visual data and achieved more than 90% accuracy on a dataset containing over 2,000 images. While these results confirm the feasibility of camera-based ASL recognition, their approach primarily relies on full image input. The present study follows a similar camera-based approach but introduces an additional landmark-extraction stage using MediaPipe. Rather than passing the complete image to the classifier, the system extracts the coordinates of detected hand landmarks and uses these numerical features as input



for a Random Forest classifier, resulting in a more streamlined recognition pipeline.

Aurangzeb et al. (2024) developed a deep learning-based hand gesture recognition system and reported accuracies of 99.23% on the Massey University Dataset and 99.00% on the American Sign Language Alphabet Dataset. These results highlight the strong performance of convolutional neural networks (CNNs) when trained on image data. However, CNNs typically require substantial computational resources, making them less suitable for real-time execution on standard devices. The present study addresses this limitation by separating hand detection from classification. MediaPipe is used to generate a compact set of landmark coordinates, which are then classified by a Random Forest model. This approach significantly reduces the amount of data the classifier must process and lowers computational demand.

Rastgoo et al. (2021) reviewed deep learning and classical machine learning approaches to vision-based isolated gesture recognition. Their analysis showed that supervised classification algorithms perform more effectively when provided with structured landmark features rather than raw pixel matrices. The authors concluded that isolated sign recognition becomes both more accurate and computationally efficient when feature extraction is decoupled from classification. This finding supports the modular architecture adopted in the present study, where MediaPipe handles landmark extraction and Random Forest performs classification.

Adithya et al. (2023) evaluated traditional machine learning models for isolated gesture recognition in human-computer interaction. Their results demonstrated that tree-based ensemble algorithms deliver reliable multiclass classification performance

on keypoint coordinate data without the high computational cost associated with deep learning models. These findings support the decision to use a lightweight machine learning approach for real-time gesture recognition.

Shin et al. (2021) developed an ASL alphabet recognition system that extracts 21 spatial joint features through hand pose estimation instead of relying on heavy image-based convolutional networks. By representing hand gestures as geometric feature vectors, their approach achieved high accuracy on static signs while remaining computationally efficient. Their findings confirm that keypoint extraction is a viable alternative to CNN-based methods. However, their work focused primarily on alphabet classification and did not integrate recognition with physical hardware control. The present study builds on this approach by combining MediaPipe keypoint extraction with a Random Forest classifier and routing recognized commands to household appliances through an Arduino-based interface.

Syulistyo et al. (2025) investigated keypoint-based gesture recognition using MediaPipe and demonstrated the feasibility of creating efficient and lightweight systems that can operate on devices with limited computational resources. Zhang et al. (2020) presented Google's MediaPipe Hands framework, which employs a two-stage pipeline consisting of palm detection followed by landmark regression to estimate 21 three-dimensional keypoints from a standard RGB video frame. Their architecture eliminates the need for depth sensors or GPU-intensive processing, making it suitable for real-time applications. The present study adopts this framework as the foundation for hand landmark extraction.

Lugaresi et al. (2019) introduced MediaPipe as a modular framework for



constructing real-time perception pipelines. The framework organizes computer vision processing into interconnected calculators connected by data streams, significantly reducing computational and memory overhead compared with traditional frame-by-frame processing. In the present study, MediaPipe is specifically used for hand landmark detection to generate structured numerical features for classification.

Sarma and Laskar (2019) demonstrated that Random Forest classifiers can achieve millisecond-level inference speeds when operating on structured numeric features rather than high-dimensional image data. Ramasamy et al. (2022) further showed that Random Forest models can automatically evaluate the importance of different spatial coordinates, allowing the algorithm to reduce feature dimensionality. Kumar and Kumar (2023) observed that Random Forest's ensemble structure helps mitigate the effects of minor variations in camera focus and keypoint detection. Gupta et al. (2024) compared tree-based ensemble algorithms with deep neural networks and found that a Random Forest classifier paired with coordinate extraction achieves comparable accuracy while requiring significantly less memory and processing power. These findings collectively justify the selection of Random Forest as the primary classification method in the present study.

Lopes et al. (2025) developed GECO, a real-time MediaPipe-based gesture controller for IoT smart home systems. Their platform supported both binary and analog gestures, operated fully on-device, and achieved latency under 50 milliseconds, with positive user evaluations. While the study successfully demonstrated the practicality of MediaPipe for real-time home automation, it focused on general hand gestures rather than structured American Sign Language

commands. The present study builds on this work by training a Random Forest classifier specifically on ASL gestures, enabling a more accessible interface designed for Deaf and Hard-of-Hearing users.

Overall, the reviewed literature supports the technical feasibility of the proposed system. Previous studies have shown that camera-based sign language recognition is achievable, that deep learning methods can attain high accuracy but often at a high computational cost, and that classical machine learning algorithms can effectively classify isolated gestures when provided with structured features. MediaPipe has been established as an efficient framework for real-time hand tracking, and existing systems have successfully applied these techniques to interactive applications. However, many studies focus either on isolated recognition without hardware integration or on general gesture control without specifically addressing ASL. The present study addresses this gap by developing a system that maps predefined ASL commands to home appliance actions using a standard webcam, MediaPipe landmark extraction, Random Forest classification, and an Arduino-based control interface.

IoT Smart Home Control and Microcontroller Hardware Integration

Traditional smart home assistants commonly rely on voice recognition and auditory feedback to allow users to interact with household devices. Although these technologies provide convenient hands-free control, voice-dependent interaction can create accessibility barriers for Deaf and Hard-of-Hearing (DHH) individuals. Sta. Maria and Deja (2024) examined the accessibility limitations of conventional voice-based smart home assistants and emphasized the importance of alternative

interaction methods that do not depend on speech recognition or audio feedback. Their study highlights the potential of non-voice interaction methods, such as gestures and visual indicators, to make smart home technologies more accessible to DHH users. This provides an important foundation for the present study because it demonstrates the need for an interaction method that allows DHH users to communicate with smart home devices without relying on spoken commands or auditory responses.

The development of accessible home automation systems also requires an appropriate combination of software and hardware. Anuradha et al. (2023) developed an IoT-based home automation system intended to support individuals with different disabilities. Their system utilized a NodeMCU microcontroller, relay modules, and multimodal interaction methods to provide users with alternative ways of controlling household appliances. The findings demonstrate that low-cost microcontrollers can be integrated with software-based interfaces to provide accessible control of electrical devices. The study also emphasizes how alternative interaction mechanisms can contribute to greater independence in operating home appliances. However, while the system supports users with disabilities, it does not specifically focus on recognizing complete American Sign Language (ASL) commands. The present study extends this concept by focusing on sign language as the primary interaction method for controlling home appliances.

One important component in developing sign language-based interaction is the ability to accurately recognize hand gestures through a camera. Yeware et al. (2023) developed and validated a sign language recognition system that captured hand

movements using a standard webcam and processed the captured visual information using a Long Short-Term Memory (LSTM) model. Their work demonstrates that a conventional camera can be used as a non-contact input device for recognizing sign language gestures without requiring users to wear specialized sensors or gloves. The use of LSTM also demonstrates the capability of deep learning to analyze sequences of hand movements, which is particularly relevant for signs that involve movement over time. However, deep-learning approaches can require greater computational resources and may be more complex to implement on systems with limited processing capabilities. This creates an opportunity to investigate lighter machine-learning approaches that can operate using extracted hand landmarks rather than processing complete image data.

An alternative approach to directly processing visual images is the extraction of hand landmarks. Yang et al. (2023) developed an IoT-based smart home control system that utilized Google's MediaPipe framework to identify hand movements. MediaPipe was used to extract 21 three-dimensional hand keypoints from the user's hand, providing structured landmark information that could be used for gesture recognition. Their system achieved a reported precision of 98.80% in recognizing the gestures used in their experiment and utilized an ESP8266 microcontroller to control household appliances. The study demonstrates that hand-landmark extraction can provide a practical approach to vision-based gesture recognition while reducing the need to process complete raw images for every classification step.

However, Yang et al. (2023) primarily relied on geometric calculations, particularly the angles between detected hand landmarks, to distinguish gestures.

Although this approach can work effectively for a limited set of predefined gestures, manually defining geometric rules may become more difficult as the number and complexity of gestures increase. More complex ASL commands may involve multiple fingers, different hand orientations, combinations of landmarks, or movements that are difficult to represent using only manually defined angle thresholds. The present study therefore builds upon Yang et al.'s use of MediaPipe for extracting 21 three-dimensional hand keypoints but replaces the reliance on manually defined geometric calculations with a Random Forest classification approach. By using the extracted landmark coordinates as features, the proposed system aims to automatically learn patterns associated with different ASL commands and classify them into their corresponding commands.

The hardware component is also essential because recognized commands must eventually be translated into physical actions. Hashim et al. (2023) developed an IoT-based home automation system using an Arduino UNO, Wi-Fi technology, and the Blynk platform to monitor and control household appliances. Their system demonstrated how a microcontroller can serve as an interface between software commands and physical electrical devices. The system allowed appliances to be monitored and controlled through a network connection, demonstrating the feasibility of integrating software-based control with household hardware. Although the system was not designed specifically for sign language interaction, its architecture demonstrates the practical use of a low-cost microcontroller for controlling appliances after receiving commands from a software interface.

When considered together, these studies establish several important

components for an accessible sign language-based smart home system. Sta. Maria and Deja (2024) establish the accessibility problem associated with voice-dependent smart home assistants and the need for alternative interaction methods. Yeware et al. (2023) demonstrate that a standard webcam can be used for sign language recognition through visual input and sequence-based machine learning. Yang et al. (2023) demonstrate that MediaPipe can extract 21 three-dimensional hand landmarks that can be used for efficient vision-based gesture recognition and smart home control. Hashim et al. (2023) demonstrate the feasibility of using low-cost microcontrollers to connect software commands with household appliances, while Anuradha et al. (2023) demonstrate the potential of accessible and multimodal home automation for individuals with disabilities.

Despite these contributions, a gap remains in combining these approaches into a system specifically designed to recognize ASL commands for smart home control using an accessible and relatively lightweight computer-vision and machine-learning approach. Existing studies either focus on accessibility and alternative interaction methods, use deep-learning approaches for sign recognition, or use simple geometric calculations for gesture-based appliance control. The present study addresses this gap by integrating webcam-based hand detection through MediaPipe with a Random Forest classifier for recognizing ASL commands and connecting the recognized commands to a home automation interface. In this approach, the webcam serves as the visual input, MediaPipe extracts the 21 three-dimensional hand landmarks, the Random Forest model classifies the extracted features into predefined ASL commands, and the recognized command is transmitted to the hardware interface for

appliance control. The proposed approach is intended to provide a more direct interaction method for DHH users by allowing hand signs to function as commands rather than relying on voice input. It also maintains the advantages of a camera-based system because users do not need to wear specialized gloves or attach physical sensors to their hands. At the same time, the use of landmark-based features and Random Forest classification provides an alternative to processing complete images through a computationally intensive deep-learning model.

Software Quality Standardization ISO/IEC and DHH Accessibility Evaluation

To evaluate the quality and accessibility of the proposed system, it is essential to go beyond technical accuracy and consider standardized quality models and design principles that prioritize the needs of the deaf. First, Fitriani et al. (2021) tested assistive software systems using the ISO/IEC 25010 standard and proposed a basis for validating real-time computer vision and IoT systems. Their study demonstrated the importance of assessing the product's functions and performance to determine its suitability for use and achieve effective operation in non-contact interfaces, which was not always considered in similar research. Thus, applying the ISO/IEC 25010 quality model to the proposed system, specifically the MediaPipe landmark extraction and Random Forest pipeline processing, ensures that the functions of the sign-language-based home automation system meet the software engineering requirements.

Then, Peres et al. (2018) explored performance efficiency, including time behavior and resource usage, in interactive real-time systems in the context of the ISO/IEC 25010 framework. According to the researchers, interfaces that demonstrate

unacceptable delays frustrate users and cause errors during interaction with computer vision systems. Similar to the previous work, this study's findings are applicable to the current project because they explain how the sign-language-driven home automation system should operate to avoid delays in processing hand landmarks detection and classification with the MediaPipe and Random Forest pipeline to ensure rapid appliance control.

Lastly, Kushalnagar and Cavender (2020) designed a set of software guidelines for Deaf and Hard-of-Hearing (DHH) people to ensure accessibility and usability. The authors stated that existing systems do not consider DHH users' communication habits and practices, forcing them to adapt their behavior when interacting with devices. They formulated recommendations for developers, including the implementation of natural Sign Language gestures and timely feedback, which helps build intuitive interfaces for DHH users. Therefore, it is possible to design the proposed gesture-controlled home automation system for people who are deaf or hard of hearing according to these accessibility guidelines to ensure intuitive control of appliances using natural Sign Language gestures.

The ISO/IEC 25010 quality model's practical implementation in different software fields supports the choice of these standards for the current research. Indeed, Canlas et al. (2021) applied the ISO/IEC 25010 framework to evaluate an information system using multiple quality characteristics, demonstrating the effectiveness of this approach. Another study conducted by Mulyawan et al. (2021) involved a literature review about the application of ISO/IEC 25010 in academic, government, mobile, and decision-support system fields and revealed that this



standard allows developers to prioritize characteristics considered critical for specific systems. Therefore, following the findings of similar research, the current study uses the ISO/IEC 25010 quality model to explore the most crucial aspects of the sign-language-driven home automation system.

Real-time sign language recognition, lightweight landmark extraction, machine learning, and IoT hardware integration come together to provide Deaf and hard-of-hearing users with an accessible, nonverbal smart home solution that gets around the voice-reliance restrictions of traditional smart assistants. In order to achieve fast, millisecond-level inference with high accuracy and built-in tolerance to spatial tracking noise, dynamic hand-landmark keypoints are extracted and classified using a Random Forest model instead of processing computationally demanding raw video frames through resource-intensive deep learning networks. Physical household appliances can switch instantly without latency bottlenecks when this keypoint feature processing and classification is offloaded to a primary computing unit. This allows for smooth signal transmission to inexpensive microcontrollers and relay modules. Through continuous, modular sprint testing, this multi-domain architecture is developed within an iterative Agile Scrum framework to accommodate the unpredictable integration requirements of physical hardware and computer vision. In order to confirm that the system satisfies strict software engineering standards while providing an empowering, user-centric interface, the final platform is methodically assessed against ISO/IEC 25010 software quality benchmarks, with an emphasis on functional suitability, performance efficiency, interaction capability, reliability, and security.

1.3 Significance Of The Study

The development of this Home Automation Sign Language Recognition System provides an accessible, non-verbal interface that translates American Sign Language (ASL) gestures into real-time appliance control. With the Google's MediaPipe coordinate extraction with an ensemble Random Forest classifier on resource-conscious hardware, the study benefits the following groups:

The Deaf and Hard-of-Hearing (DHH) Community: Offers a visual alternative to mainstream voice-activated systems, enabling users to independently control household appliances with greater autonomy and ease.

Caregivers and Family Members: Lowers caregiver dependency by allowing DHH individuals to manage routine environmental controls independently through an intuitive interface.

IT Students and Researchers: Serves as a practical reference for pairing landmark extraction with lightweight machine learning models to prevent processing bottlenecks on standard consumer hardware.

Assistive Technology Developers: Demonstrates how to design responsive, cost-effective assistive tools verified against the ISO/IEC 25010 software quality.

IoT and Embedded Systems Integrators: Provides a working implementation for serial communication and command execution between host-based machine learning pipelines and low-cost microcontrollers.

Future Researchers: Establishes a baseline architecture for integrating continuous gestures, two-handed tracking, and wireless communication protocols into smart home systems.

1.4 Statement Of The Problem

The general objective of this study is to design, develop, and evaluate a real-time Sign Language recognition system using MediaPipe and a Random Forest algorithm to control home appliances through a Smartphone Camera, Arduino Uno R3 microcontroller, and relay module via the IoT.

Specifically, this study seeks to determine the following:

1. What are the stages undertaken in developing the Home Automation Sign Language Recognition with application of MediaPipe Using Random Forest Algorithm?.
2. How MediaPipe and the Random Forest algorithm are used in developing the Home Automation Sign Language recognition?
3. What is the software quality level of the system as evaluated by IT experts and Deaf and Hard-of-Hearing (DHH) users based on the ISO/IEC 25010 standard in terms of:

   4.1 Functional Suitability

   4.2 Performance Efficiency

   4.3 Interaction Capability

   4.4 Reliability

   4.5 Security
4. What is the implementation plan in the developed Home Automation Sign Language Recognition with application of MediaPipe Using Random Forest Algorithm.

1.5 Scope and Limitations Of The Study

The study will concentrate on the general operation of the Home Automation Sign Language Recognition System, which will identify specific American Sign Language (ASL) gestures for controlling household appliances using MediaPipe and the Random Forest Algorithm. The system's capacity to identify static gestures and translate them into corresponding commands will be the main focus of the study. The hand gestures will be recorded using a Smartphone, and 21 three-dimensional hand landmarks will be extracted from the images by MediaPipe. The Random Forest Algorithm will then be used to process the extracted data locally on a laptop in order to identify the chosen gestures. A 5V relay module that is connected to five household appliances will be controlled by an Arduino Uno r3 microcontroller that receives the recognized commands using serial or wireless communication. Accuracy, affordability, real-time performance, and a few ISO/IEC 25010 criteria, such as Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security, will all be assessed. The study will take place in a controlled indoor setting from November 2026 through the first semester of the 2026–2027 academic year.

It is important to recognize that the research will have a number of limitations. First, even though MediaPipe will extract 21 hand landmarks, the system will only identify five static ASL gestures for controlling appliances. As a result, the system will not support continuous ASL sentence translation, the full ASL dictionary, or the entire ASL vocabulary. Second, the system will only be tested in a controlled indoor setting, which might not accurately reflect real-world circumstances like varying lighting,

backgrounds, camera angles, distances, and hand occlusion. The system will only be available to users who can execute the necessary hand gestures; people with severe visual impairments or those who are unable to execute the gestures may find it inappropriate.

The system will be designed specifically for local, real-time appliance control and will not require an internet connection or cloud-based processing. In this setup, a smartphone serves as the primary visual sensor to capture the live video feed using a wired USB connection. A laptop handles all core computations, using MediaPipe for feature extraction and a Random Forest model for gesture classification. The laptop transmits commands over USB serial to an Arduino Uno R3, which acts as the microcontroller driving the relay modules connected to household appliances. Only five home appliances and simple switching operations, like turning appliances on and off, will be supported by the system. A mobile application, a dedicated GPU, multi-hand recognition, or full ASL translation will not be included in the study. These restrictions will limit the system's use to the chosen hardware configuration, appliances, gestures, and controlled testing environment.

Chapter II

Methodology

This chapter acts as the blueprint for the research study, outlining the systematic methodology used to achieve the research objectives. It bridges system-level software allocation and software design, allowing for the clear specification of software functions, performance requirements, and constraints. In addition, it demonstrates how to construct models of the data, functional, and behavioral domains that the software will address.

2.0 Project Overview

Project Information & Overview

The project is an innovative home automation concept that enables Deaf, hard-of-hearing, and speech-impaired people to communicate with their home appliances. The technology uses a regular webcam to track hand movements through the MediaPipe library and uses a Random Forest classifier with 21 three-dimensional coordinates representing the key points of the hand. In addition, the project uses a microcontroller connected to relay modules that close or open circuits of selected devices, such as lights and electric fans.

The system works by allowing the user to make gestures in front of a webcam that are captured and analyzed in real-time. First, the live video feed is processed by the MediaPipe library that recognizes 21 landmarks as a set of points that describe the exact location of coordinates. Next, the Random Forest algorithm classifies these coordinates and compares it to a database of known American Sign Language

commands. Once a command is recognized, it is converted into a set of instructions for the microcontroller to activate selected devices through relay modules.

Requirements Definition

a. Business (Core function of the system)

The system should fulfill its primary business function of hand gestures made by the user into commands that turn selected home appliances. The technology should consistently and accurately perform the task after detecting a command closing or opening circuits of home appliances. Finally, the system should work reliably under regular lighting conditions with a regular webcam for visual tracking of movement.

b. User

This project is primarily designed for the Deaf, hard-of-hearing, and speech-impaired community that means that its primary functions should cater to their specific needs and potential limitations. In this case, the most important design considerations are the system's ability to reliably recognize and differentiate hand gestures. In addition, the user should be able to operate the system intuitively with a few smart home appliances. Other potential users of the technology are people who care for those who cannot speak or hear, such as family members and caregivers. These users should be able to view the information the program provides and operate the system without extensive experience with technology.

c. Security

The system should be designed in a way that prevents unauthorized access and interference with its regular operations. Hand gestures should only trigger system commands and thus home appliance activation if the user initiates the action. The security element is essential because the system processes video input in real-time, eliminating the possibility of denying access once the authentication is successful. A potential vulnerability is the connection between the recognition software and the microcontroller that operates the relay modules, meaning that an unauthorized third party could intercept video data transmitted to the microcontroller. A regular user should only have limited access to the system, such as changing the default password or adjusting the commands that trigger specific system responses.

2.1 Concept of the study

The architectural framework presents the overall structure of the proposed Home Automation Sign Language Recognition System and illustrates how each component works together to control household appliances.

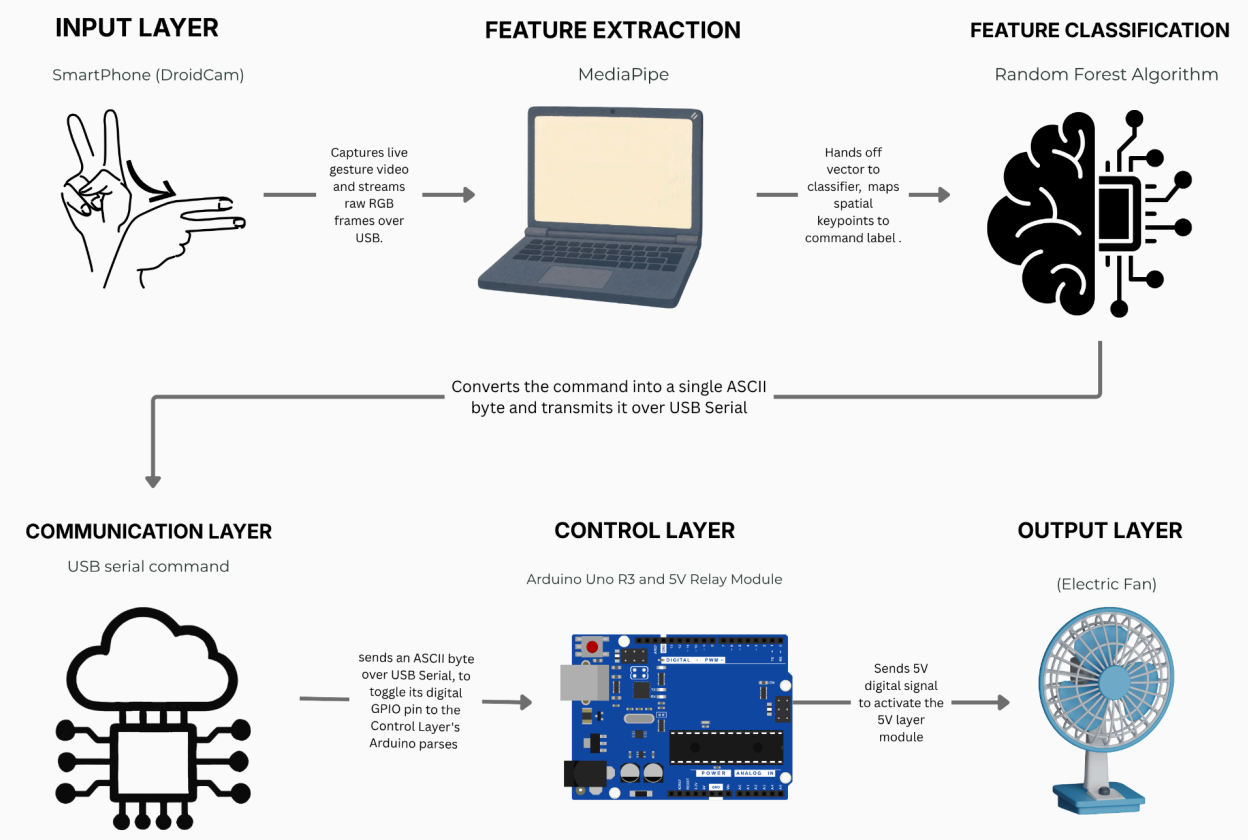


Figure 1.1 Architectural Diagram

shows the flow of data from gesture input to appliance control, highlighting the relationship between the system's input layer, feature extraction, feature classification, communication layer, control layer, and output layer.

INPUT LAYER The application uses the smartphone camera as a video input device which captures user hand gestures. USB is used to communicate the RGB live video stream to the next stage as the primary source of visual input to the system.

FEATURE EXTRACTION The application takes single video frames from the input layer, recognizes the hand and finally extracts 21 three-dimensional coordinates representing the current position of the fingers. As a feature vector, the application thus uses 63 numbers representing the x, y, and z coordinates of each detected landmark.

FEATURE CLASSIFICATION The application receives the coordinates detected in the previous processing step and classifies them according to the trained American Sign Language gestures. Since the feature vector is a 63-dimensional array, classification takes less than ten milliseconds to process and complete. The output of this stage is a command which defines the recognized gesture.

COMMUNICATION LAYER The application encodes the command into a single ASCII byte and sends it via USB serial communication from the laptop to the Arduino Uno R3 microcontroller. This layer provides reliable data transmission between the laptop and the microcontroller.

CONTROL LAYER The application receives the ASCII command transmitted from the communication layer via USB Serial and subsequently activates the appropriate digital GPIO pin. The activated pin energizes a 5V relay which controls the targeted device in the home appliance control scenario.

OUTPUT LAYER The application turns the targeted device (electric fan) on or off depending on the received command. In this manner, the application closes the gesture recognition loop and completes the processing chain.

2.2 Conceptual operation

This section outlines the system's operation using Process Flowchart. The Flowchart illustrates how data moves and transforms through each functional process.

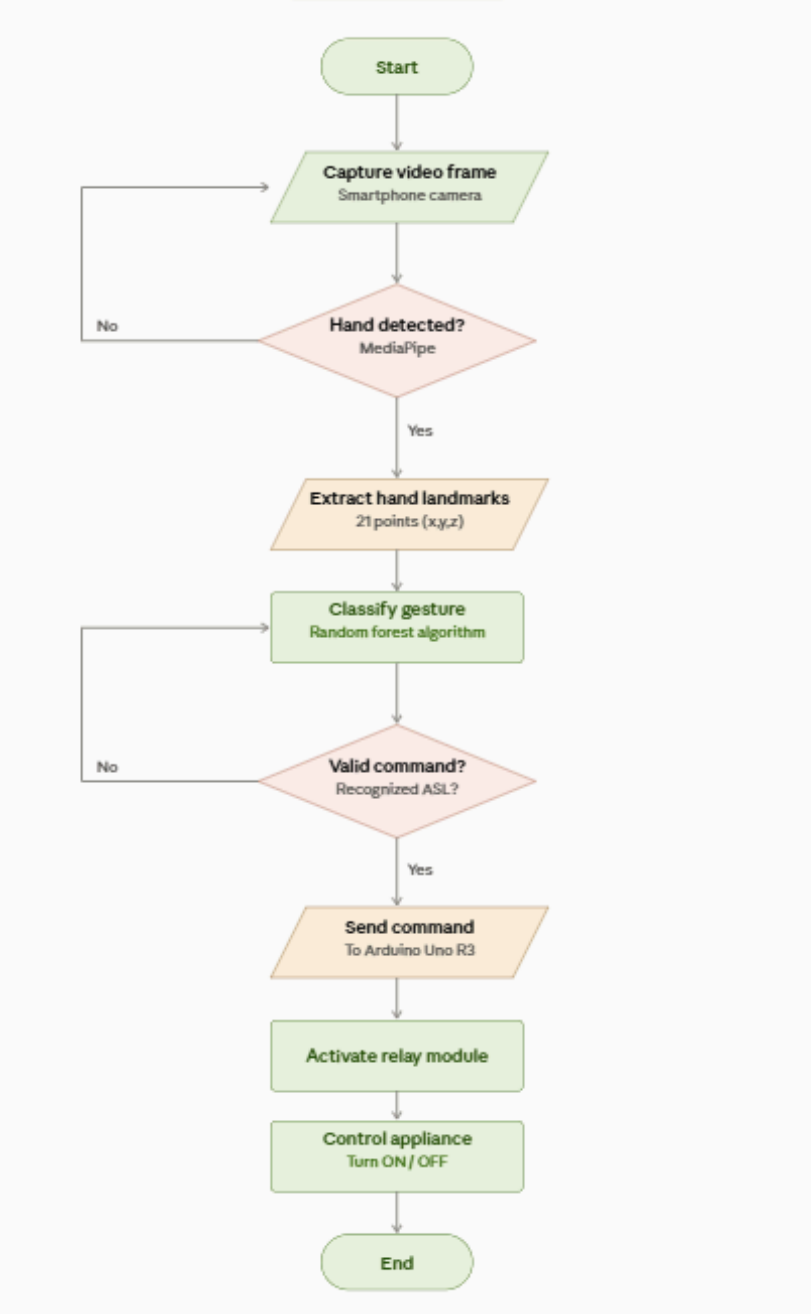


Figure 1.2 Process Flowchart

Shows how a recorded video frame is validated and processed through stages of detection, extraction, classification, and command verification before starting appliance control. This illustrates the sequential logic flow of the gesture recognition system. Feedback loops ensure that only appropriate gestures result in an action.

The proposed system's architecture consists of interconnected functional layers that process user gestures into physical actions. Each layer serves a distinct operational purpose, working together to provide Deaf, Hard-of-Hearing (DHH), and speech-impaired individuals with an accessible, real-time gesture-driven home automation interface.

The Sensing and Input Layer acts as the primary data acquisition boundary that allows users to interact with the system. It uses a smartphone camera connected via USB to a host laptop through the DroidCam app to capture and stream live video frames of user hand gestures.

The Computer Vision and Machine Learning Layer serves as the computational core running on the host laptop. Within this layer, Google's MediaPipe framework extracts 21 three-dimensional spatial keypoints (63 spatial coordinates per frame) and verifies real-time hand presence. A Random Forest classifier receives these structured coordinate arrays and classifies them against pre-established static American Sign Language (ASL) commands. Decoupling keypoint extraction from decision classification guarantees rapid processing speed without requiring high-performance graphics processing units.

The Communication Layer manages data serialization and signal transmission between the host laptop and the microcontroller. Once an ASL command is verified,

this layer converts the instruction into ASCII command byte and transmits it over a wired USB Serial interface to ensure low latency and reliable data delivery.

The Hardware Control and Actuation Layer executes the physical switching of household appliances. The digital General Purpose Input/Output (GPIO) pins of an Arduino Uno R3 microcontroller trigger a 5V relay module, which acts as an electromechanical switch connected to a high-voltage alternating current (AC) load circuit. Upon receiving the serial command, the microcontroller drives the relay to open or close the circuit, successfully switching the target appliance (such as an electric fan) ON or OFF to complete the end-to-end operation in real time.

2.3 Mockups

This section presents the visual design and user interface of the system, illustrating how users interact with its core features.

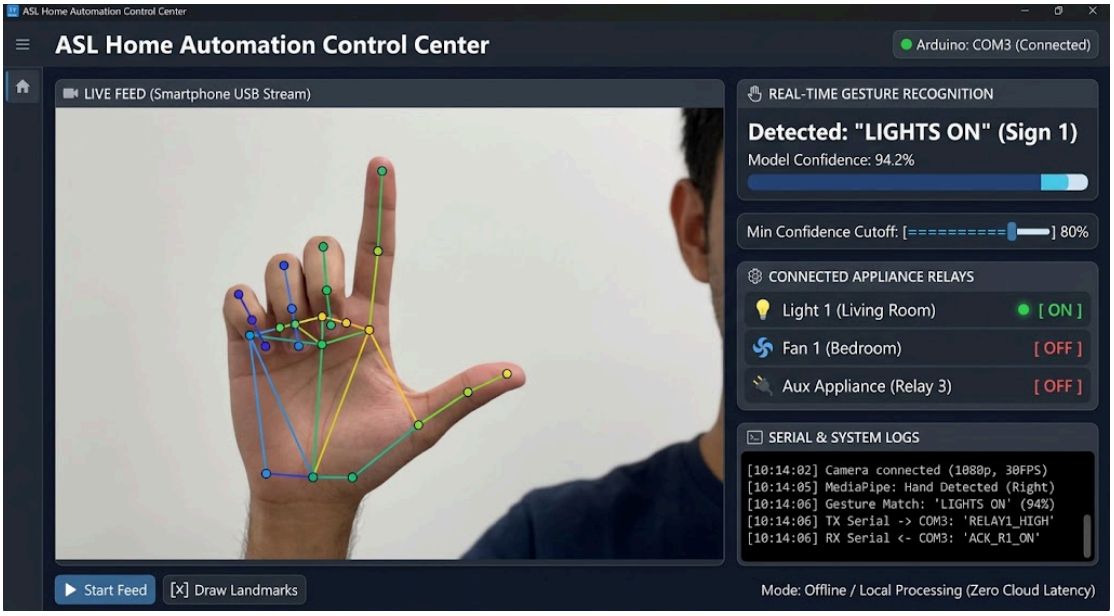


Figure 1.3. user interface

illustrates the software architecture and edge-computing execution pipeline for the ASL Home Automation system.

UI Layout Description:

Header Bar: Establishes the project title and provides a critical "Arduino: COM3 (Connected)" status badge, reassuring the panel that the hardware link is active.

Live Feed (Left): Shows the video source from the smartphone, overlaid with MediaPipe landmarks. This visualizes the feature extraction stage of your methodology.

Real-Time Gesture Recognition (Right-Top): Displays the output of the Random Forest classifier, including the predicted sign and the confidence score. It also includes a slider for setting a "Min Confidence Cutoff" (e.g., 80%) to manage system accuracy.

Connected Appliance Relays (Right-Middle): The most crucial part for a hardware project—a visual dashboard showing the actual state of your physical relays ([ON]/[OFF]).

Serial & System Logs (Right-Bottom): Provides transparency by showing the raw bidirectional communication between the laptop and the Arduino, proving that a gesture was recognized and a command was physically sent.

Footer: Emphasizes your local, offline operational mode, aligning directly with your requirement for no internet/cloud reliance.

2.4 Methodology of the study

Rationale: Agile software development using the Scrum framework was selected because of the project's multi-layered architecture, which combines computer vision, machine learning algorithms, and physical electrical circuits. Developing these interrelated components concurrently requires high flexibility. An iterative, sprint-based approach allows the team to train, test, and recalibrate hand-tracking models and serial communication protocols independently without altering the physical relay wiring. Furthermore, conducting incremental testing between low-voltage digital logic and high-voltage AC loads helps identify and resolve integration bottlenecks early, ensuring that the interface continuously adheres to the non-verbal accessibility requirements of Deaf and Hard-of-Hearing (DHH) users.

Phases:

1. Requirements Analysis & Sprint Planning
2. Iterative Software & ML Development
3. Hardware-Software Integration
4. System & Accessibility Testing
5. System Deployment & Finalization

Implementation:

Requirements Analysis & Sprint Planning (1 Week):

During the initial week, the team establishes the project baseline by defining DHH accessibility requirements and UI visual feedback standards. Key operational

specifications are set, including a 30 FPS camera input, mappings for five static ASL gestures, an 80% confidence threshold, and Arduino Uno R3 GPIO pin allocations.

Iterative Software and Machine Learning Development (2 Weeks)

Sprint 1: Configure the video acquisition pipeline using OpenCV with a smartphone camera via DroidCam over USB, and integrate Google's MediaPipe Hands to extract 21 three-dimensional hand landmarks (63 coordinate features).

Sprint 2: Compile the keypoint dataset, train and tune the Random Forest classifier, implement confidence threshold filtering, and establish pySerial ASCII byte encoding.

Hardware-Software Integration (2 Weeks)

Sprint 3: Develop Arduino C++ firmware for non-blocking serial command parsing, wire the optoisolated 5V relay module to the designated GPIO pins, and test the physical actuation of targeted AC loads (lighting and electric fan).

System and Accessibility Testing (1 Week)

Sprint 4: Conduct comprehensive end-to-end trials to evaluate gesture classification accuracy across varied indoor lighting, track pipeline latency (MediaPipe extraction, inference duration, and relay triggering), and evaluate visual indicator responsiveness.

System Deployment and Finalization (1 Week) The concluding week is dedicated to packaging the complete software-hardware system and finalizing project documentation. The microcontroller, relay module, and associated wiring are assembled and mounted securely within a protective physical enclosure to ensure

operational safety during live demonstrations. In parallel, the local desktop application is packaged for standalone execution, comprehensive user documentation is authored, and final revisions to the research manuscript are completed.

Benefits:

Modular Decoupling: Isolates software classification updates from physical hardware, allowing model retraining without requiring circuit redesigns.

Early Risk and Fault Mitigation: Incremental verification across low-voltage signals and mains-level electrical loads ensures safe hardware operation and isolates serial communication faults early in the lifecycle.

User-Centric Refinement: Regular sprint reviews allow for iterative adjustments to visual UI dashboards and confidence thresholds, ensuring the application remains intuitive and fully accessible for DHH individuals without relying on auditory cues.

Resource Optimization: Facilitates rapid benchmarking of lightweight algorithms, ensuring the recognition pipeline maintains real-time performance on standard consumer laptops.

2.4.2 Software Quality Evaluation Framework

To evaluate the overall software quality level of the developed system, an empirical evaluation protocol based on the ISO/IEC 25010 Software Quality Model is administered. This evaluation determines whether the gesture-driven automation setup meets established software engineering and Interaction Capability standards beyond model accuracy alone.

Respondents and Evaluators

The evaluation will be administered to two target respondent groups to collect both technical and accessibility data:

I**T Experts (N = 3):** Software engineers, systems analysts, and embedded systems developers who evaluate the system's technical architecture, specifically Functional Suitability, Performance Efficiency, Reliability, and Security.

Deaf and Hard-of-Hearing (DHH) Users (N = 5): Primary end-users who assess the Interaction Capability, visual dashboard clarity, and the accessibility of the non-verbal gesture controls.

Evaluation Instrument and Scoring

Data will be gathered using structured survey questionnaires scored on a 5-point Likert scale (5 = Excellent, 4 = Good, 3 = Fair, 2 = Poor, 1 = Very Poor), combined with automated hardware/software performance logs. The evaluation covers the five ISO/IEC 25010 characteristics outlined in the study's objectives:

Functional Suitability: Measures the functional completeness and correctness of the system in recognizing the five defined static ASL gestures and activating the designated appliance relay without command misrouting or omission.

Performance Efficiency: Assesses computational behavior and time efficiency, specifically measuring MediaPipe extraction latency, Random Forest prediction duration (targeting <10 ms), serial transmission response time, and local CPU and RAM utilization on the host computer.

Interaction Capability: Measures interface comprehensibility, ease of

performing the selected gestures, readability of visual status widgets (relay states and confidence score bars), and overall operational autonomy for DHH users without requiring acoustic signals.

Reliability: Measures system fault tolerance and operational consistency under continuous operation, tracking immunity to key point jitter, error handling during video frame drops, and avoidance of false triggers.

Security: Evaluates transmission integrity over the local USB serial line, validation of user-initiated gesture triggers, and the privacy safeguards provided by operating in an entirely offline, local execution environment.

References

Adithya, V., Rajesh, R., & Kumar, S. (2023). Machine learning approaches for vision-based isolated gesture recognition in interactive systems. Journal of Visual Communication and Image Representation, 90, 103721. https://doi.org/10.1016/j.jvcir.2022.103721

Akbar, M. A., Mahmood, S., Shafiq, M., Alsanad, A. A., & Gumaei, A. (2022). Empirical evaluation of Agile implementation challenges in Internet of Things (IoT) software development. IEEE Access, 10, 43812–43828. https://doi.org/10.1109/ACCESS.2022.3168812

Anuradha, P., Vasanth, K., Renuka, G., & Rajeshwar Rao, A. (2023). IoT based enabling home automation system for individuals with diverse disabilities. e-Prime—Advances in Electrical Engineering, Electronics and Energy, 6, Article 100366. https://doi.org/10.1016/j.prime.2023.100366

Aurangzeb, K., Javeed, K., Alhussein, M., Rida, I., Haider, S. I., & Parashar, A. (2024). Deep learning approach for hand gesture recognition: Applications in deaf communication and healthcare. Computers, Materials & Continua, 78(1), 127–144. https://doi.org/10.32604/cmc.2023.042886

Banag, C. T. (2023). Development of sign language tutorial mobile application for Filipinos. International Journal of Research in Education Humanities and Commerce, 4(2), 124–132. https://doi.org/10.37602/IJREHC.2023.4213

Bosch, J., Olsson, H. H., & Crnkovic, I. (2021). Engineering AI systems: A research agenda. Journal of Systems and Software, 181, Article 111050. https://doi.org/10.1016/j.jss.2021.111050

Canlas, R. B., Piad, K. C., & Lagman, A. C. (2021). An ISO/IEC 25010 based software quality assessment of a faculty research productivity monitoring and prediction system. In 2021 9th International Conference on Information Technology: IoT and Smart City (ICIT 2021). ACM. https://doi.org/10.1145/3512576.3512619

Fitriani, E., Suzianti, A., & Purnomo, H. D. (2021). Quality assessment of assistive user interfaces using the ISO/IEC 25010 software quality model. Journal of Software Engineering and Applications, 14(8), 415–429. https://doi.org/10.4236/jsea.2021.148024

Grande, A. M., Ayala, R. E., Isteefano, S., & Mohd, T. K. (2023). Implementing computer vision techniques to recognize American Sign Language (ASL) hand signals. Procedia Computer Science, 218, 2023–2032.

Gupta, R., Sharma, M., & Singh, V. (2024). Lightweight machine learning frameworks for IoT-enabled smart home gesture control interfaces. Internet of Things, 25, 101042. https://doi.org/10.1016/j.iot.2023.101042

Hashim, M. a. A., Fauzi, M. a. M., Zin, M. F. M., & Noh, K. S. S. K. M. (2023, April 30). IOT-based home automation system using Arduino Uno. https://journal.uitm.edu.my/index.php/JOA/article/view/5086

Hron, M., & Obwegeser, N. (2022). Why and how is Scrum being adapted in practice: A systematic review. Journal of Systems and Software, 183, 111110. https://doi.org/10.1016/j.jss.2021.111110

Hutchinson, C., Cleland, J., Williams, P. A. H., Manuel, K., & Laver, K. (2024). Calculating the social impact of home automation for people with disability: A social return on investment study. Australian Occupational Therapy Journal, 71(6), 956–966. https://doi.org/10.1111/1440-1630.12977

Klopp, M., Röpke, R., & Schroeder, U. (2020). Scrum in higher computer science education: A systematic literature review. Proceedings of the 15th International Conference on Computer Science & Education (ICCSE), 451–456. https://doi.org/10.1109/ICCSE49874.2020.9201382

Kumar, A., & Kumar, P. (2023). Robust vision-based sign language recognition using keypoint landmarks and ensemble tree classifiers. Computers and Electrical Engineering, 108, 108689. https://doi.org/10.1016/j.compeleceng.2023.108689

Kushalnagar, R., & Cavender, A. (2020). Designing accessible software interfaces for Deaf and Hard of Hearing users: Accessibility guidelines beyond text captions. ACM Transactions on Accessible Computing, 13(2), 1–22. https://doi.org/10.1145/3386321

Lopes, M. C., Silva, P. A., Marenco, L., Vilas Boas, E. C., Carvalho, J. G. A. d., Ferreira, C. A., Carvalho, A. L. O., Guimarães, C. V. R., Aquino, G. P., & de Figueiredo, F. A. P. (2026). GECO: A real-time computer vision-assisted gesture controller for advanced IoT home system. Sensors, 26(1), 61. https://doi.org/10.3390/s26010061

Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Zhang, F., Chang, C.-L., Yong, M. G., Lee, J., Chang, W.-T., Hua, W., Georg, M., & Grundmann, M. (2019). MediaPipe: A framework for building perception pipelines. arXiv preprint arXiv:1906.08172. https://doi.org/10.48550/arXiv.1906.08172

Mulyawan, M. D., Kumara, I. N. S., Swamardika, I. B. A., & Saputra, K. O. (2021). Kualitas sistem informasi berdasarkan ISO/IEC 25010: Literature review. Majalah Ilmiah Teknologi Elektro, 20(1), 15–24. https://doi.org/10.24843/MITE.2021.v20i01.P02

Peres, S. C., Pham, T., & Phillips, R. (2018). Validation of software usability and performance efficiency metrics under ISO/IEC 25010 in real-time user interfaces. International Journal of Human-Computer Interaction, 34(11), 1012–1023. https://doi.org/10.1080/10447318.2017.1418805

Ramasamy, P., Gopalakrishnan, S., & Baskar, S. (2022). Feature selection and classification of sign language landmarks using Random Forest ensemble models. Pattern Recognition Letters, 158, 114–122. https://doi.org/10.1016/j.patrec.2022.04.018

Rastgoo, R., Kiani, K., & Escalera, S. (2021). Hand sign language recognition using computer vision and machine learning techniques: A survey. Expert Systems with Applications, 164, 113794. https://doi.org/10.1016/j.eswa.2020.113794

Sarma, D., & Laskar, F. A. (2019). Real-time hand gesture recognition using Random Forest ensemble classifier for human-computer interaction. Journal of Visual Communication and Image Representation, 64, 102632. https://doi.org/10.1016/j.jvcir.2019.102632

Schneider, C., Lauenroth, K., & Knauss, E. (2020). Agile project models in software engineering capstone courses: An empirical evaluation. IEEE Transactions on Education, 63(3), 182–191. https://doi.org/10.1109/TE.2019.2958741

Schwaber, K., & Sutherland, J. (2020). The Scrum Guide: The definitive guide to Scrum: The rules of the game. Scrum.org. https://scrumguides.org/scrum-guide.html

Shin, J., Matsuoka, A., Hasan, M. A. M., & Srizon, A. Y. (2021). American Sign Language alphabet recognition by extracting feature from hand pose estimation. Sensors, 21(17), 5856. https://doi.org/10.3390/s21175856

Silva, L. F., Matalonga, S., & Santos, M. (2023). Agile methodologies in embedded IoT systems development: A systematic mapping study. Journal of Systems and Software, 198, Article 111602. https://doi.org/10.1016/j.jss.2023.111602

Sta. Maria, T. J., & Deja, J. A. (2024). Alexa, I wanna see you: Envisioning smart home assistants for the Deaf and hard-of-hearing. arXiv. https://arxiv.org/abs/2412.00514

Subramanian, V., Raj, P., & Kumar, S. (2021). Adapting Agile software development practices for artificial intelligence and machine learning projects. Information and Software Technology, 139, Article 106658. https://doi.org/10.1016/j.infsof.2021.106658

Syulistyo, A., Tanaka, Y., Pramanta, D., Fuengfusin, N., & Tamukoh, H. (2025). Low-cost computation for isolated sign language video recognition with multiple reservoir computing. PLoS ONE, 20(7), e0322717. https://doi.org/10.1371/journal.pone.0322717

Tran, N., DeVries, P. S., Seita, M., Kushalnagar, R., Glasser, A., & Vogler, C. (2024). Assessment of sign language-based versus touch-based input for Deaf users interacting with intelligent personal assistants. Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 1–15. https://doi.org/10.1145/3613904.3642094

Yang, C.-Y., Lin, Y.-N., Wang, S.-K., Shen, V. R. L., Tung, Y.-C., Shen, F. H. C., & Huang, C.-H. (2023). Smart control of home appliances using hand gesture recognition in an IoT-enabled system. Applied Artificial Intelligence, 37(1), 2176607. https://doi.org/10.1080/08839514.2023.2176607

Yeware, R., Gokhale, A., Chalse, S., & Mante, J. (2023). American Sign Language detection using deep learning. 2023 7th International Conference on Computing, Communication, Control and Automation (ICCUBEA). https://doi.org/10.1109/ICCUBEA58933.2023.10392208

Zhang, F., Bazarevsky, V., Vakunov, A., Tkachenka, A., Sung, G., Chang, C.-L., & Grundmann, M. (2020). MediaPipe Hands: On-device real-time hand tracking. arXiv preprint arXiv:2006.10214. https://doi.org/10.48550/arXiv.2006.10214